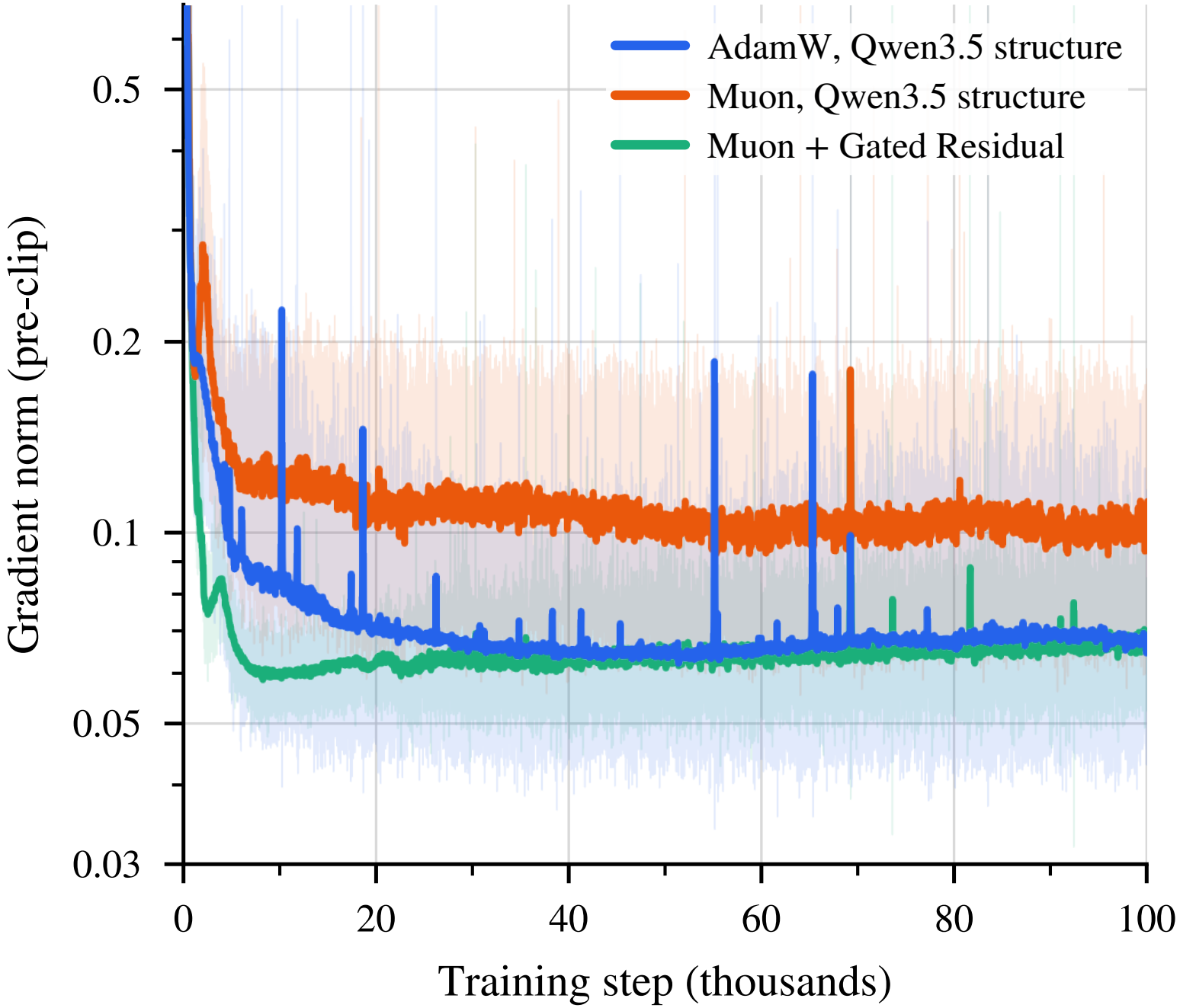

AdamW, Qwen3.5 structure
Muon, Qwen3.5 structure
Muon + Gated Residual
Gradient norm (pre-clip)
0.5
0.2
0.1
0.05
0.03
0
20
40
60
80
100
Training step (thousands)